

Carnegie Mellon University

Data Analysis and Machine Learning

What is this course about?

Adding Value with Data

Raja Sooriamurthi, Ph.D
Information Systems Program
raja@cmu.edu

Carnegie Mellon University
HeinzCollege
INFORMATION SYSTEMS • PUBLIC POLICY • MANAGEMENT



Our schedule

Lecture Course-2026 Spring			
	ET Time (12 hrs time difference)	China Time	
Lecture 1	6-8AM on Saturday, March 14	6-8PM on Saturday, March 14	
Lecture 2	6-8AM on Saturday, March 21	6-8PM on Saturday, March 21	
Lecture 3	6-8AM on Saturday, March 28	6-8PM on Saturday, March 28	Homework 1. Please send questions to TA before Lecture 3.
Lecture 4	6-8AM on Saturday, April 4	6-8PM on Saturday, April 4	Homework 2. Please send questions to TA before Lecture 4.
Lecture 5	6-8AM on Saturday, April 11	6-8PM on Saturday, April 11	
Mid-term (Specific Time TBD, on the weekend of April 18/19)			Please send questions to TA before April 13.
Lecture 6	6-8AM on Saturday, April 25	6-8PM on Saturday, April 25	
Lecture 7	6-8AM on Saturday, May 2	6-8PM on Saturday, May 2	
Lecture 8	6-8AM on Saturday, May 9	6-8PM on Saturday, May 9	Homework 3. Please send questions to TA before Lecture 8.
Lecture 9	6-8AM on Saturday, May 16	6-8PM on Saturday, May 16	Homework 4. Please send questions to TA before Lecture 9.
Lecture 10	6-8AM on Saturday, May 23	6-8PM on Saturday, May 23	
Final Term (Specific Time TBD, on the weekend of May 30/31)			Please send questions to TA before May 25.

Research part of the course

- Two options

1. Two weeks, in person, in Guangzhou (July 20 – 31)



2. Online, once a week, June 5 – August 16 (Term A)

Today's Session

- The rise of data
- Case studies in EDA, ML, BD
- Course Learning Objectives, Structure, Assessment

*Data is the new cyber-currency;
companies rely on it to optimize customer experience and drive sales —
hackers target and monetize the same data.
... Imperva*

*Data is the new oil.
It's valuable, but if unrefined it cannot really be used.
It has to be changed into gas, plastic, chemicals, etc.
to create a valuable entity that drives profitable activity;
so must data be broken down, analyzed for it to have value.
... Clive Humby*



Five Questions:

- **What** are we going to do in this course?
- **Why** are we studying it?
- **How** are we going to study it?
- **Who** are you?
- **Who** am I?

Professor Raja's Background



India $\Rightarrow$ US $\Rightarrow$ India $\Rightarrow$ US



- B.E (CS & Cengr); MS (CS); PhD (AI/CogSci)

- Engineering / Science / Business / IS

Why not?



Why?



What?



- Teaching

- Intro (Pascal, C, C++, Java, Scheme, .net)
- Data structures, discrete math
- Programming languages
- OOAD
- XML and Web technologies
- Systems Development
- Artificial Intelligence
- Intelligent Decision Support Systems
- Database Systems
- Data Science and Big Data
- Puzzles, Games, and Problem Solving
- Several awards for distinguished teaching

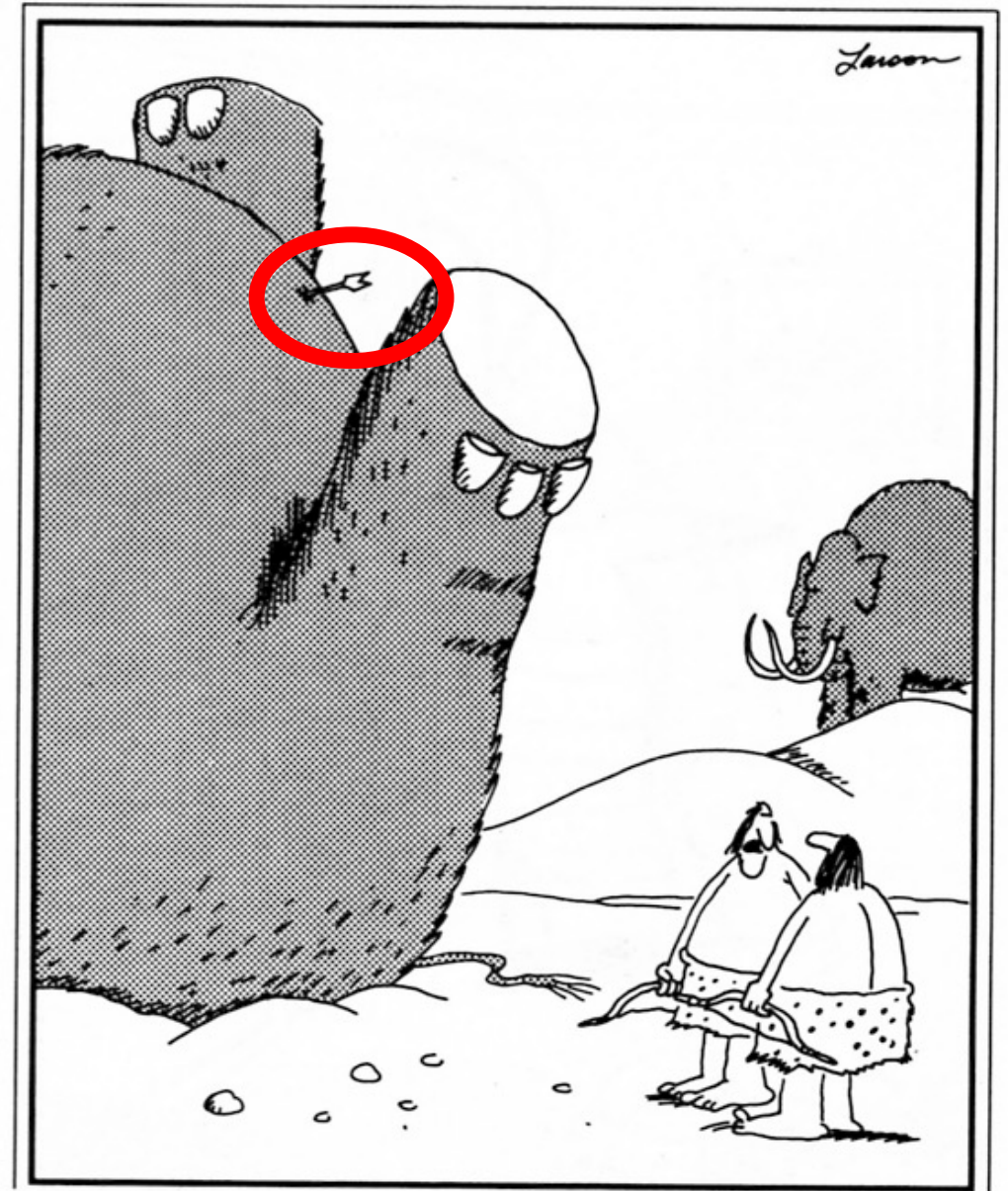
Raja Sooriamurthi

Teaching Professor, Information Systems Program

Research & Outreach

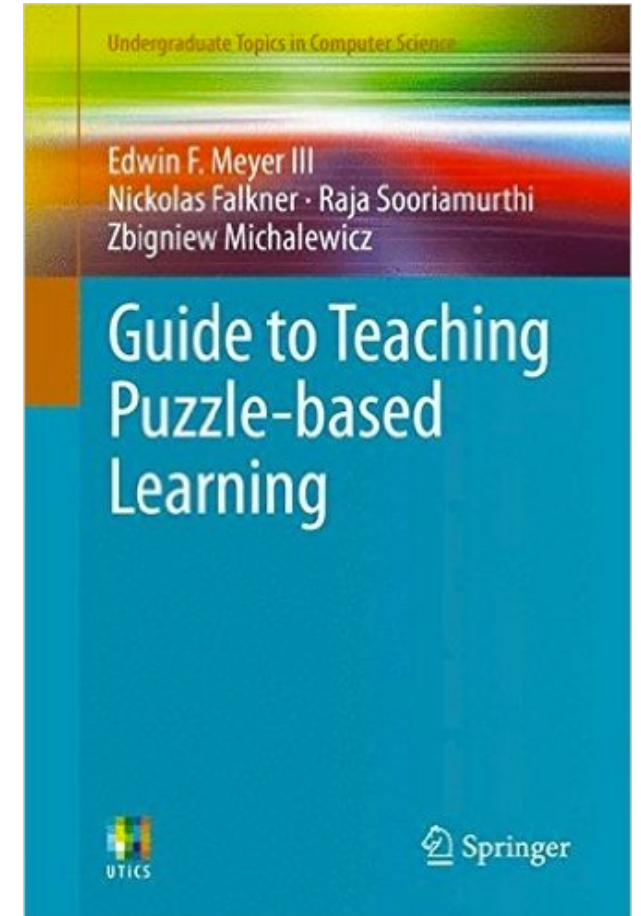
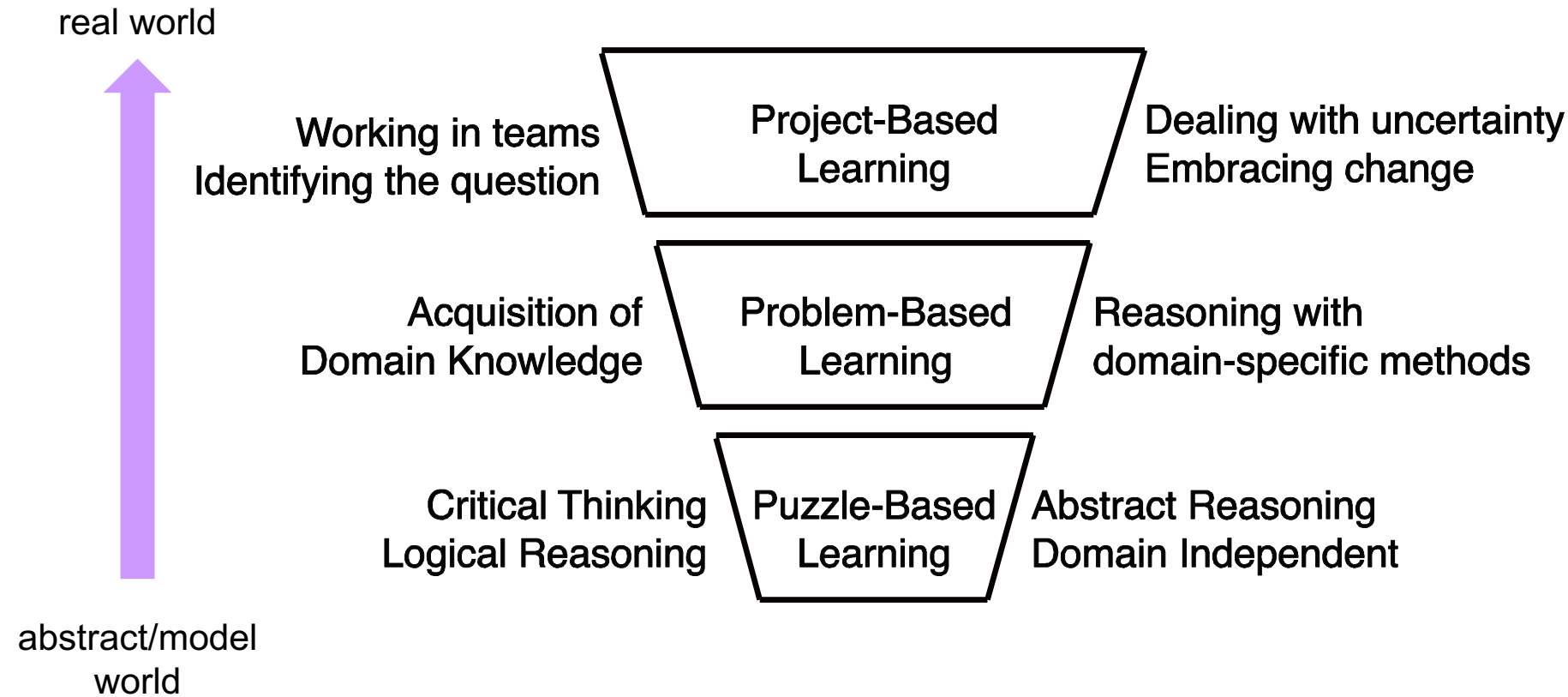
- Reasoning and Learning in Programs and People
- AI / case-based reasoning / Machine Learning

- Java Engagement for Teacher Training
- Data Jam
- STEAM workshops
- Osher Life long learning institute
- Toastmasters
- Faculty advising (CIS club, Resident Fellow)



“Maybe we should write that spot down.”

Real World Problem Solving



{Puzzle, Problem, Project}-based learning

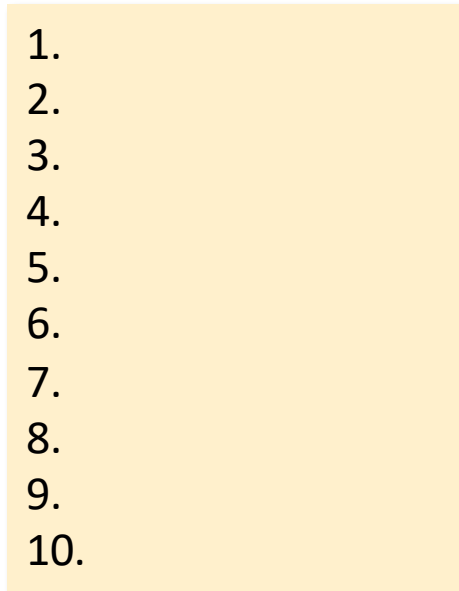
The End Game: A Puzzle with a Purpose

Name a body part that has three letters in it

(no 'naughty' words, slangs, or abbreviations)

A Puzzle with a Purpose

- Name 10 three letter body parts
 - (no 'naughty' words, slangs, or abbreviations)



1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

For 2 minutes try to solve this individually.
Then, if needed, pair up and group solve.

The Parts ...

1.

2.

3.

4.

5.

6.

7.

8.

9.

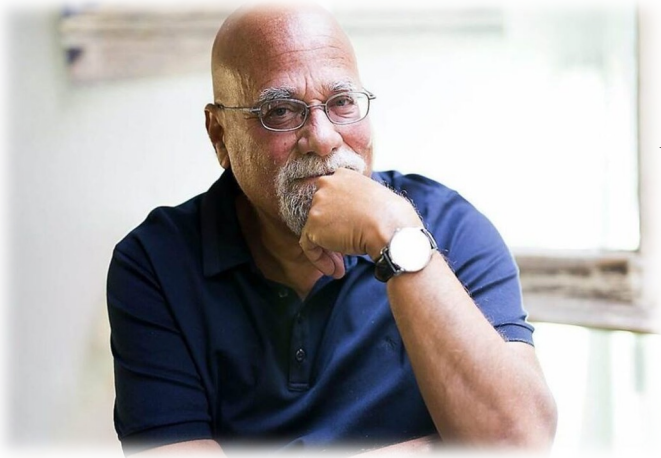
10.

What is the take away?

Sometimes we miss the obvious

Two experiences

Some thoughts leaders in AI and Education



Roger Schank (1946-2023)

Learning occurs when *someone wants to learn*, not when someone wants to teach.

Learning results from *what the student does and thinks*, and only from what the student does and thinks. The teacher can advance learning only by influencing the student to learn.

Herbert Simon (1916-2001)



Getting the most out of this Course*

Spring 2025

Raja Sooriamurthi

*I'm not a great programmer;
I'm just a good programmer with great habits.*

... Kent Beck

Software development thought leader
and the creator of the eXtreme Programming (XP) agile methodology.

As with many things in life, you will get out of this course that which you put into it. Following are some thoughts that may help you get the most out of this experience. Most significantly inculcate a **professional mindset**. There is an interesting causal chain of **Mindset → Thoughts → Actions → Results**. A professional mindset, is part of the general notion of a growth mindset. It is multi-pronged and involves:

- **Be present.** Attend class both physically and mentally. If we meet online on Zoom, please **keep your video on**. Participate (ask, share, ...) in class (attendance will be taken). As the saying goes "Eighty percent of success is just showing up". It is so tempting and easy to multi-task. Do your best not to multi-task; be grounded in the *now*.
- **Be on time.** Connected to the above, we will have a running start to the semester and each class. I subscribe to the mindset "If you are on time, you are late". Aim to be in the class room a few minutes before so that we can start class on time with full attendance.
- **Be prepared.** Complete pre-class exercises. Review the readings, slides for each class. Even a quick peruse of the slides before class will help you get "**primed**" for that session's discussion. Plan hydration and bio-break needs ahead—avoid exiting and entering while class is in session.
- **Actively participate.** Engage the instructor and fellow classmates. Any insights from the pre-class exercise? Questions from the slides? What have you read in the popular press or otherwise that is relevant to our course? What would you like to discuss in class? **Share your thoughts, experiences**.
- **Initiative, teamwork, and pride.** A definition of initiative is "doing what needs to be done without being told to do it". Own your work. Take pride in your work. Be an effective team player.

In addition to the above, ponder the following to get the most out of any work you do:

- **Good work is rework.** Get a head start on all deliverables (assignments, projects). Look into the *Pomodoro technique*. Several excellent resources on this simple but effective time management and productivity technique are available on the web. For more work strategies I'd recommend reading about Cal Newport's notion of "Deep Work". There is a lot about this on the web. A nice place to start is this short [9 minute video summarizing Deep Work](#).
- **Style matters.** When writing SQL conform to Simon Holywell's [SQL style guide](#). When writing R conform to the [Tidyverse style guide](#). When writing Python, **inculcate the Pythonic way**. Sometimes, even though one may be using the Python language, one may be "thinking in Java". Python supports several powerful idiomatic ways of expressing computation known as the *Pythonic way*. I've written about this in detail in a separate [document](#).

[Wikipedia](#) article about this book is a good place to start. [YouTube](#) has several videos on themes from this book including videos of Covey himself.

- **Have fun!**



Jupyter notebooks, a good text editor (emacs, vim, atom, sublime, vscode, ...) and the For R coding we will use Rstudio. If you don't have a preferred editor yet, [Visual](#) is a popular choice. For each tool, enhance productivity by being comfortable with minimizing the use of the GUI interface.

Join and read piazza. Make good use of the office hours of the course staff. As needed, meetings to discuss anything with us in detail.

And follow your passion in the projects. Cultivate pride in your work. Pick projects you can discuss with friends and recruiters.

At one word IS is all about: *value*. Always be sure you understand why we are doing something. If you don't agree, if you have a different point of view, please do speak up.

Embody the traits of enthusiasm, curiosity, respect, and humility to aid my learning. A professional mindset, is the foundation for everything. I have also got a lot of value out of Steven Covey's *Effective People*. It is well worth the investment to think about these ideas. The

What is Information Systems about?

If you were to boil down what the field of Information Systems is **about** to just one word, what would it be?

If you say 'X', iteratively ask yourself “why X” ? [Wikipedia: [5 whys](#)]

To what *prime directive* does this lead?

What is Information Systems about?

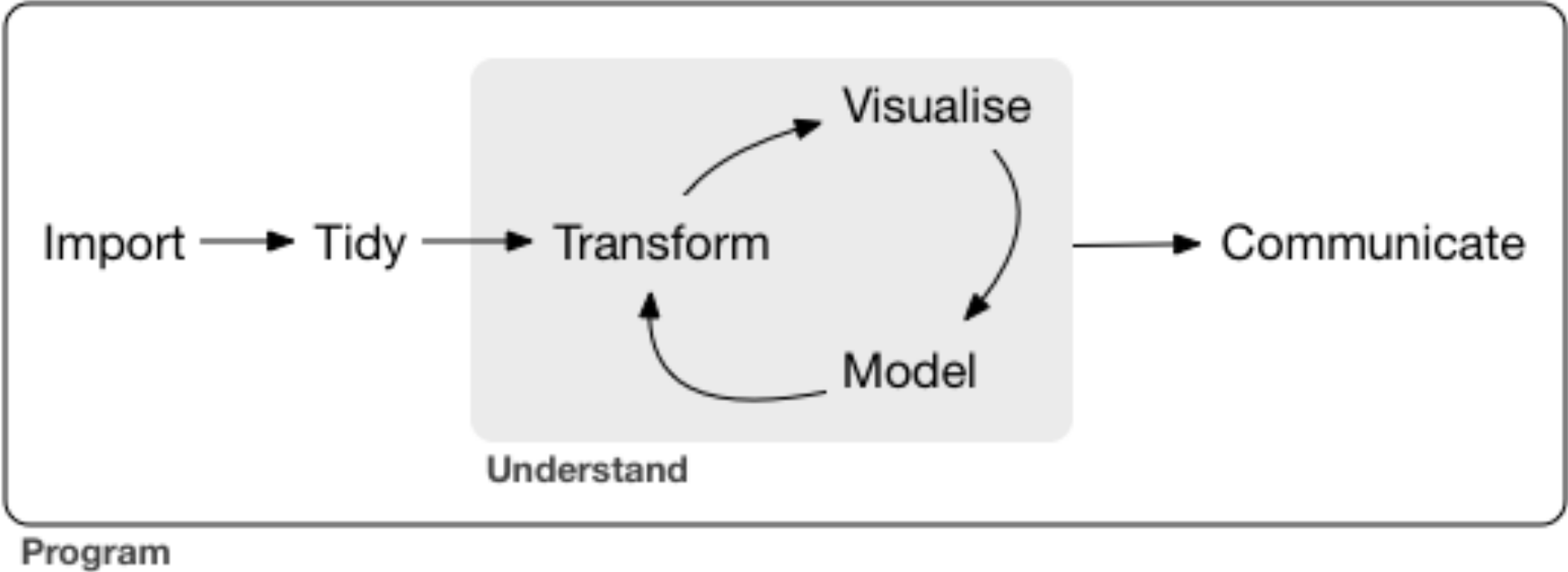
Adding

Value

To Organizations

Using Technology

Data

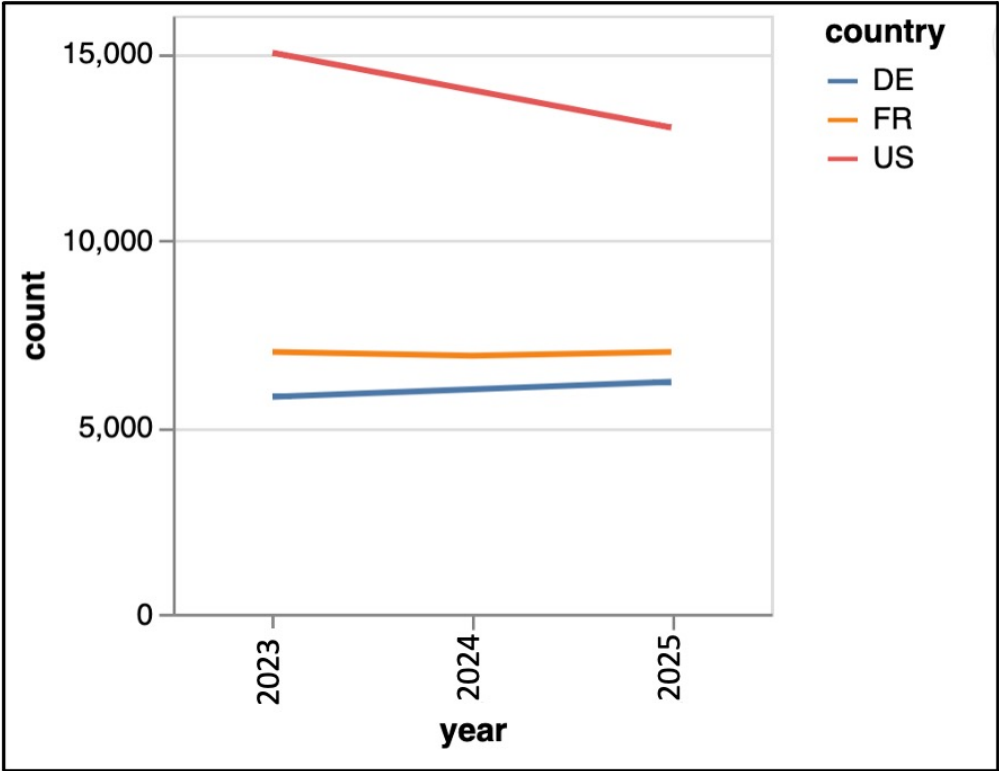


Value

Two uses of Data ... Patterns and Predictions

countries				
	country	2023	2024	2025
0	FR	7000	6900	7000
1	DE	5800	6000	6200
2	US	15000	14000	13000

visualize
patterns



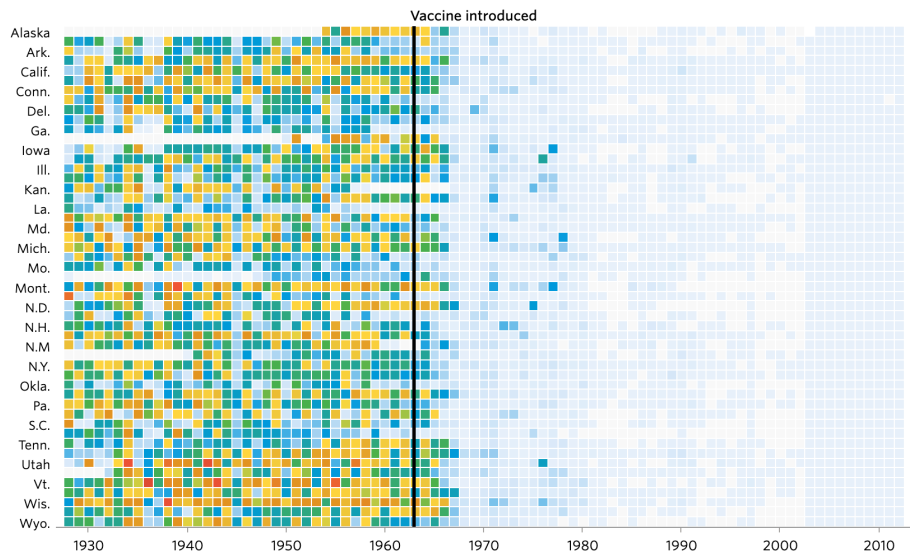
analyze
reason / learn / understand

Predict:
US / 2026 ?

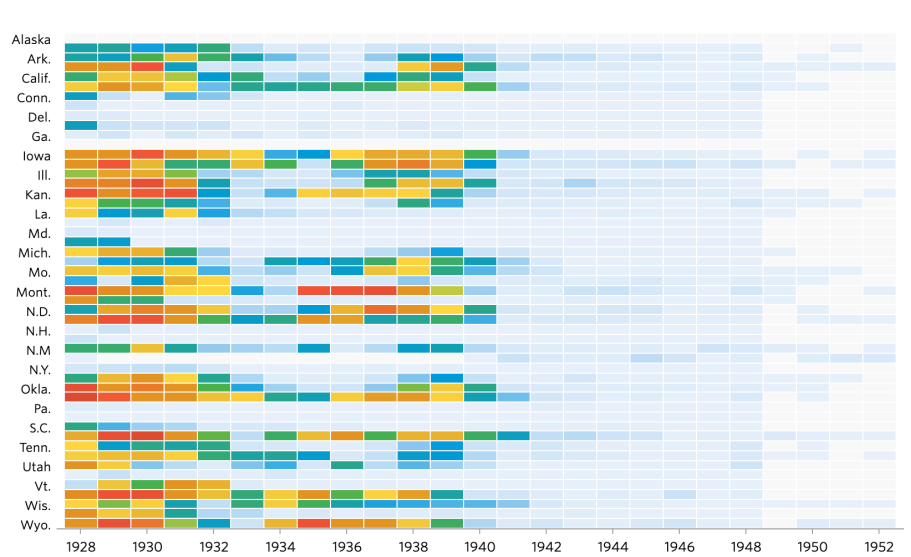
Battling Infectious Diseases in the 20th Century: The Impact of Vaccines

<http://graphics.wsj.com/infectious-diseases-and-vaccines/>

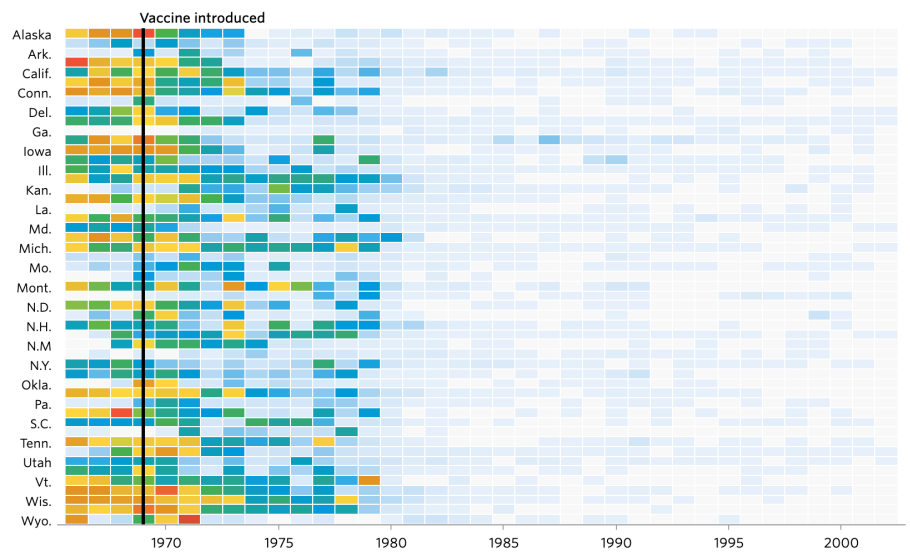
Measles



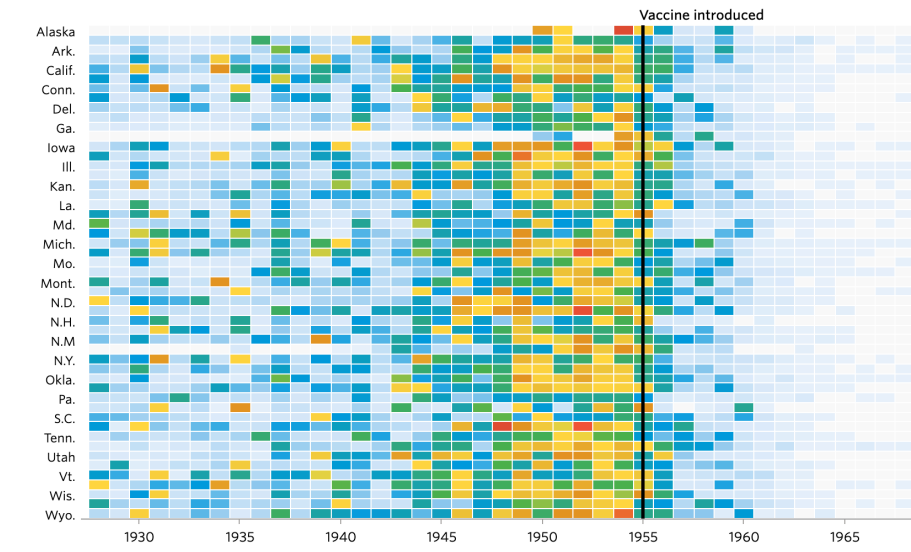
Smallpox



Rubella



Polio



THE RISE OF DATA

Why and how has data become so hot?

The Age of Data

- We are generating It
 - Generators + Sensors
- We are collecting it
- We are storing it
 - 4 TB @ \$70 (¢1.75 / GB)
- We are processing it



at&t



CDC 7600

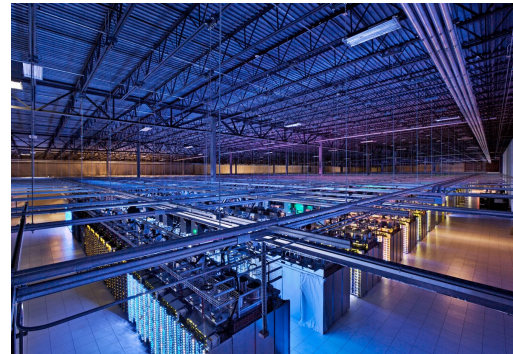
\$5M in 1976 (\$32)

vs.

\$150



iPhone4



A Google server warehouse

Seagate 4TB Terascale HDD SATA 6Gb/s
64MB Cache 3.5-Inch Internal NAS Hard
Drive (ST4000NC001) - 3 Year Warranty

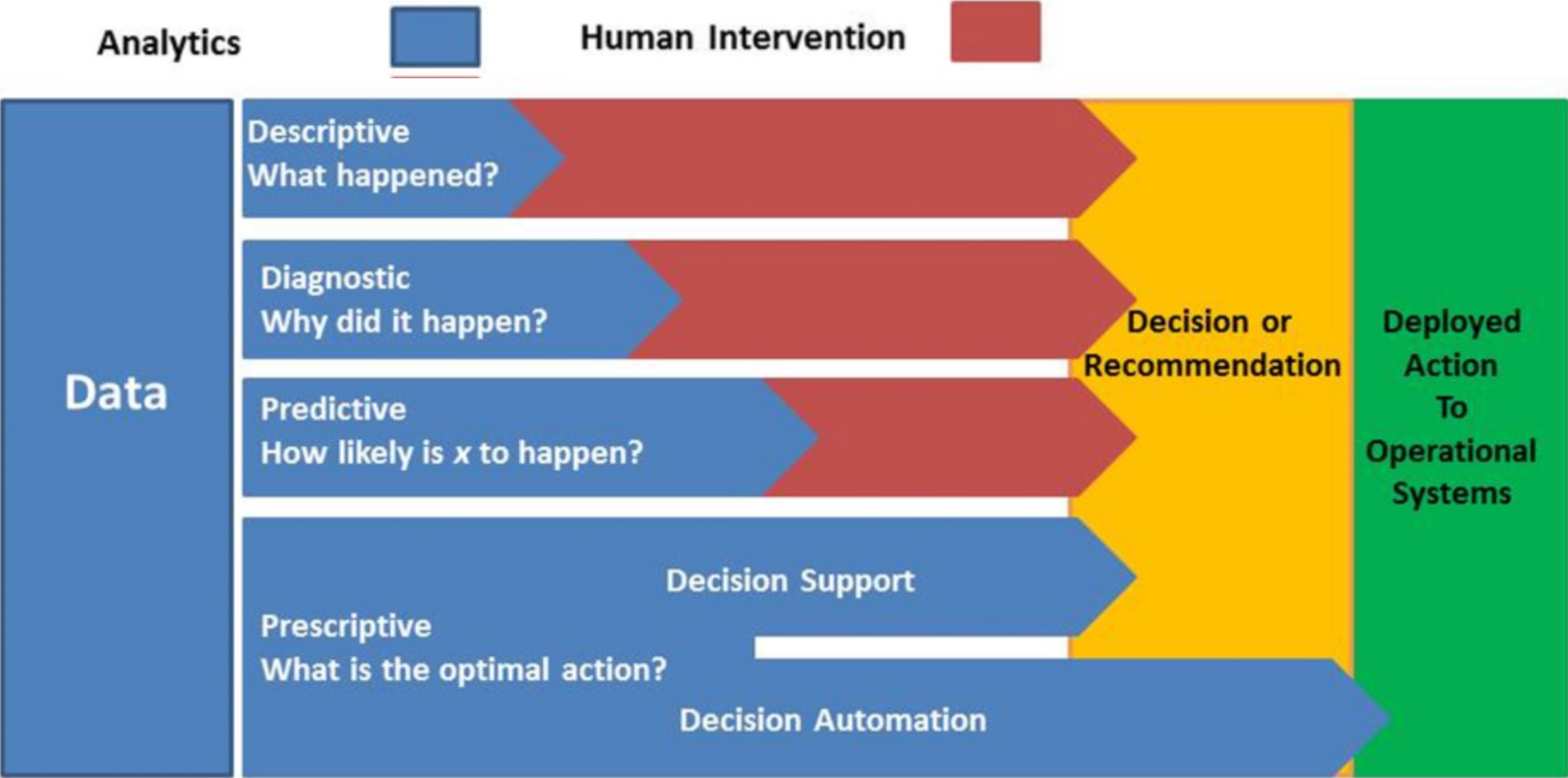
[Visit the Seagate Store](#)

★★★★★ 67 ratings | 3 answered questions

Price: **\$69.99** & **FREE Shipping**



Four Types of Analytics



CASE STUDIES

1. *Exploratory Data Analysis*
2. *Machine Learning*
3. *Big Data*

Case Studies

- EDA
 - Taxi tips
 - Old faithful eruptions
- ML
 - Pharmacovigilance
 -
- BD
 - Project Artemis
 - Watson

Book with lots of case studies of Big Data

- Book has 1 page descriptions of how companies leveraged AI, ML, BD
- File:/Users/raja/l/bd/intro

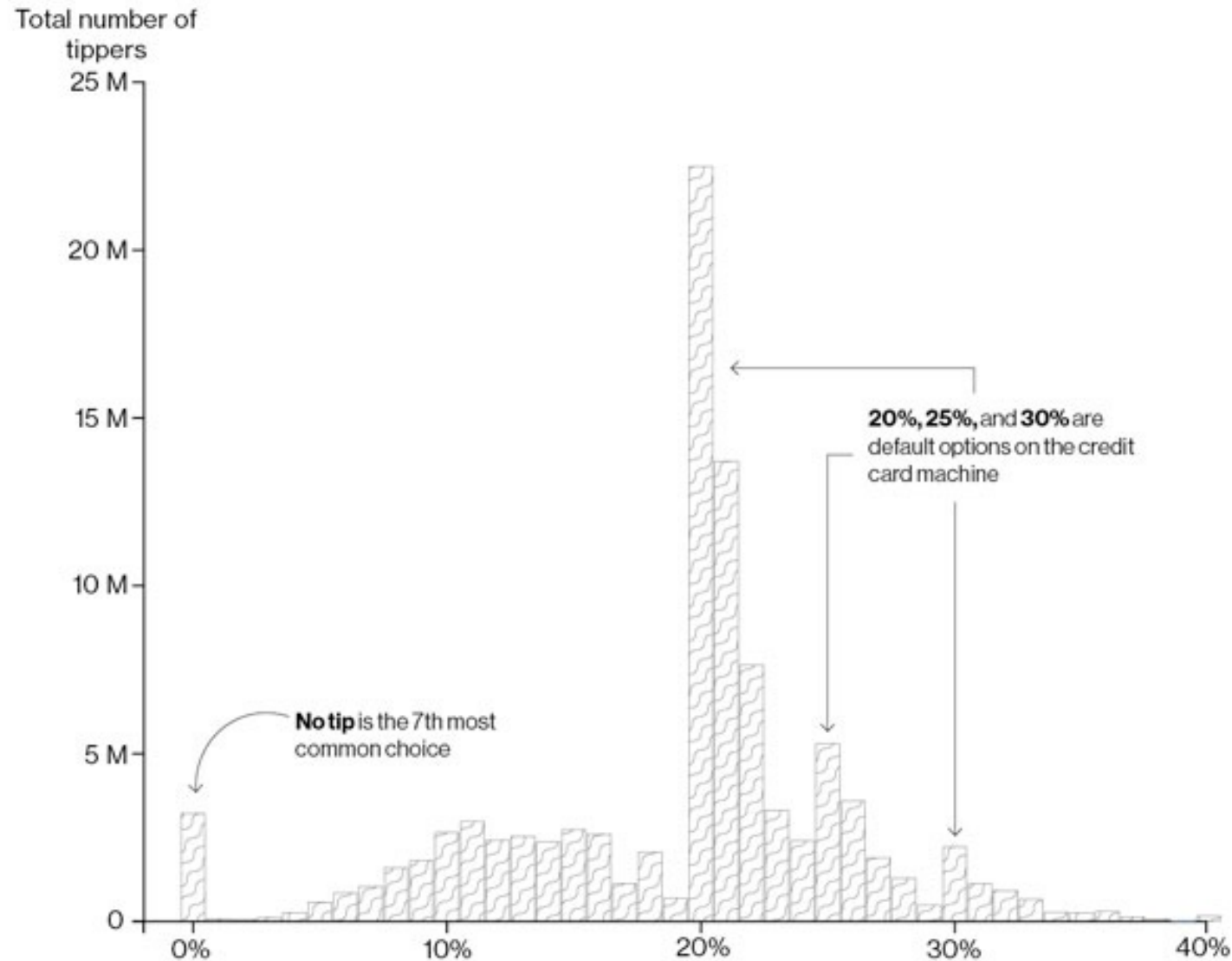
VISUALIZATION AND EXPLORATORY DATA ANALYSIS

The purpose of computing is insight, not numbers.

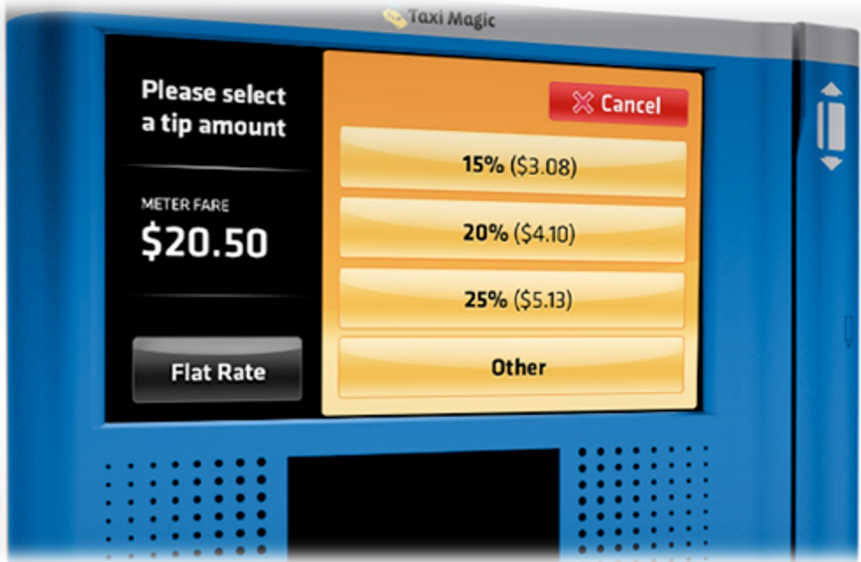
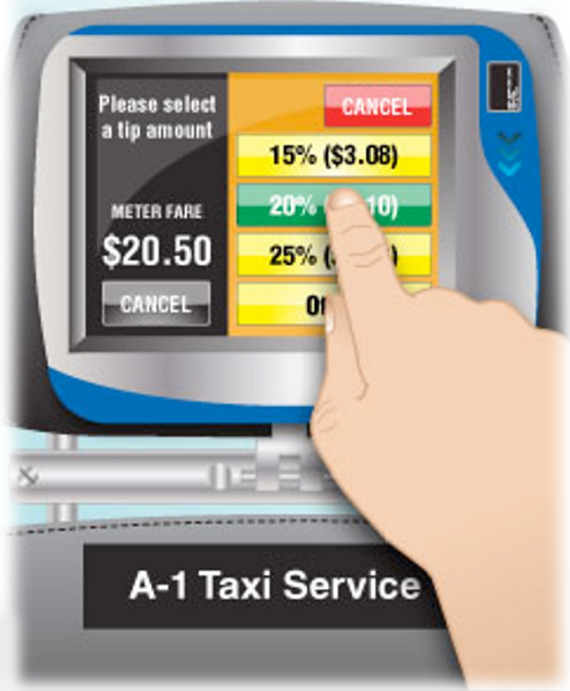
... Richard Hamming (1968, Turing award recipient)

EDA / Case Study 1: How much do people tip cab drivers?

People Most Often Tip 20-25 Percent



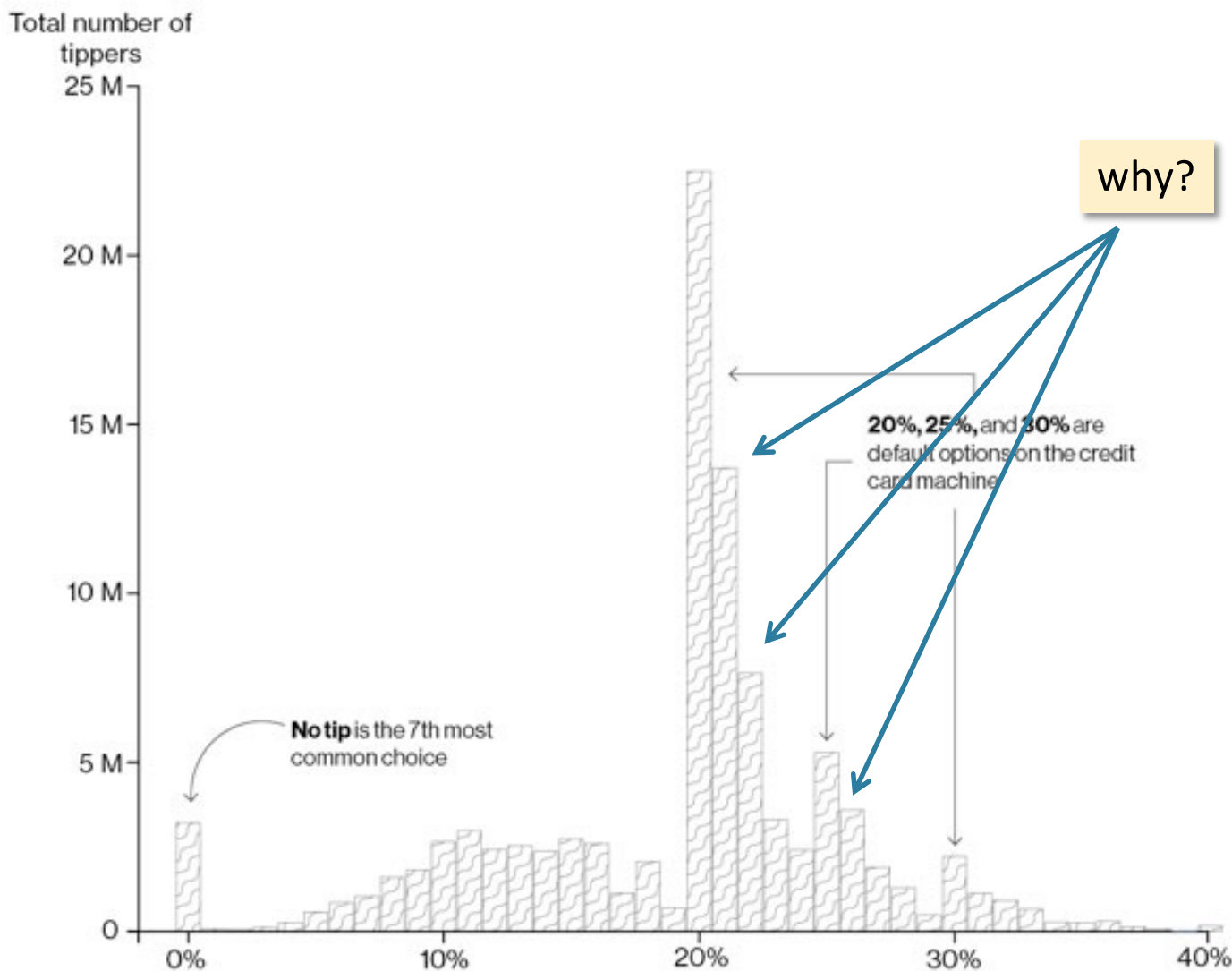
EDA / Case Study 1: Payment Meters nudge you ...



EDA / Case Study 1: What are the sibling bars about?

“The greatest value of a picture is when it forces us to notice what we never expected to see.”
— John Tukey

People Most Often Tip 20-25 Percent



Lets Explore this Data Set

- Please open the Jupyter notebook

1. medallion

2. hack_license

3. vendor_id

4. payment_type

5. fare_amount

6. surcharge

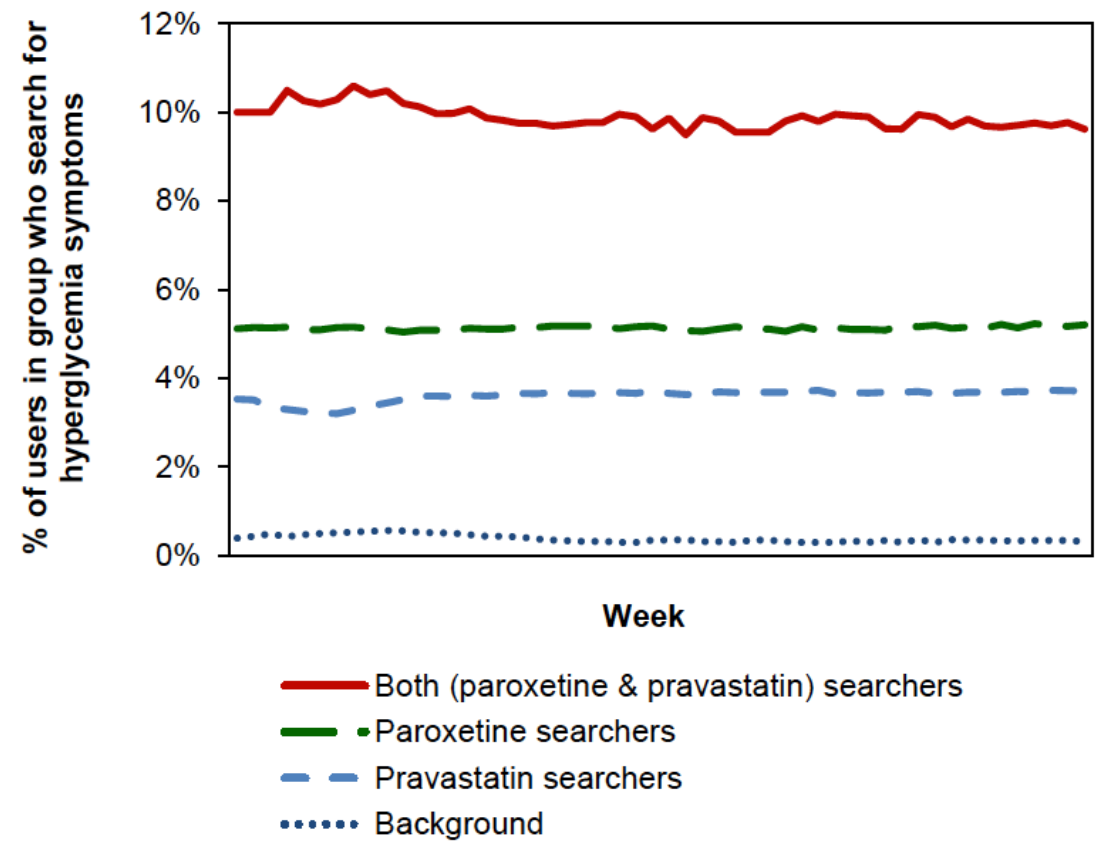
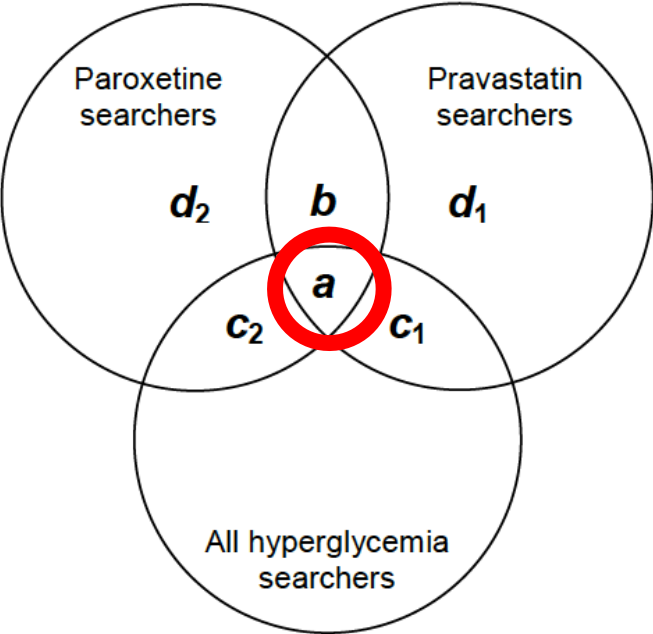
7. mta_tax

8. tip_amount

9. tolls_amount

10. total_amount

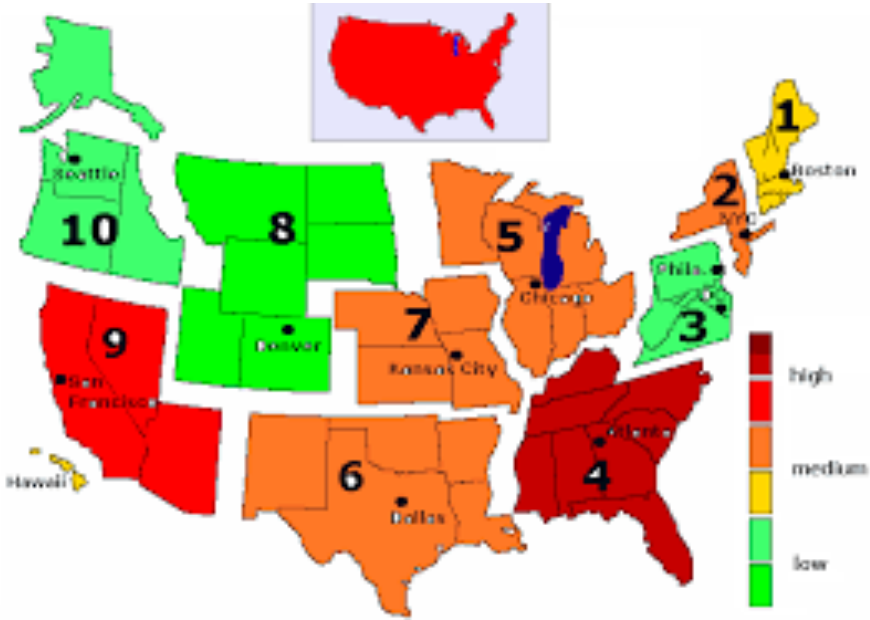
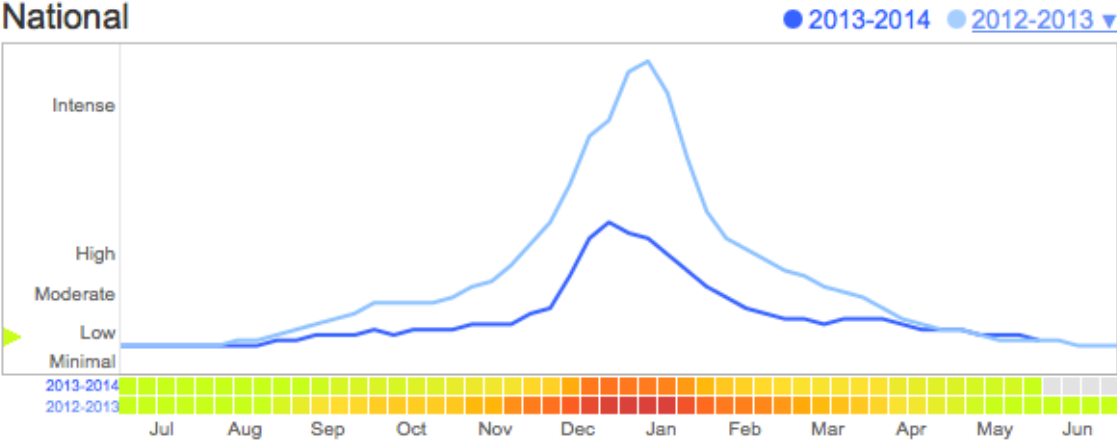
EDA / Case Study 2: Pharmacovigilance



EDA / Case Study 2: Google Flu Trends



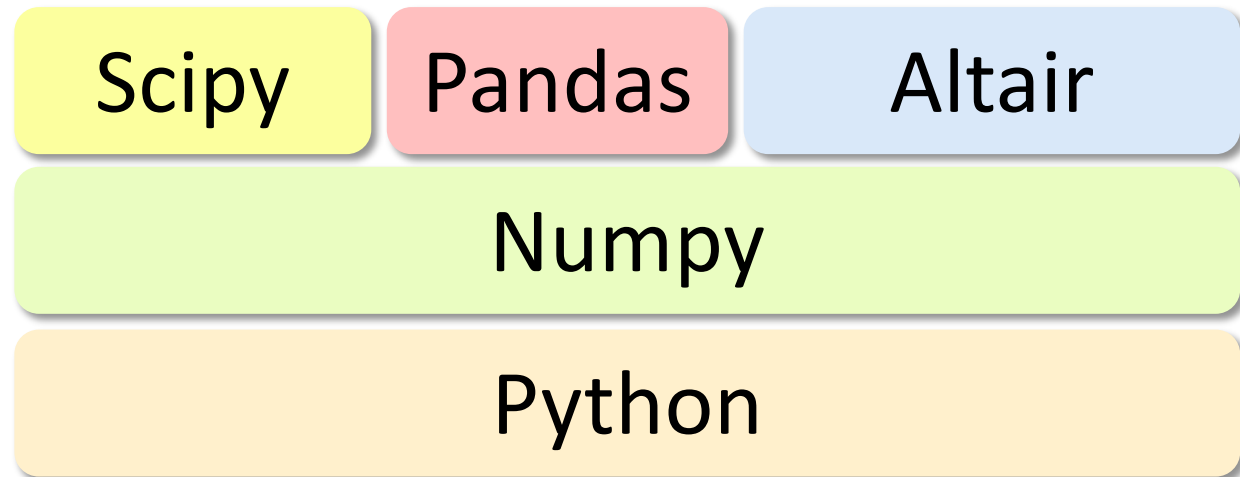
National



[Influenza Forecasting with Google Flu Trends](#)

Tools for Explorator Data Analysis (EDA)

- Python, Jupyter, Pandas
- Visualization
 - Grammar of Graphics
 - Plotnine, Altair, Plotly
- Web Scrapping
 - BS4, Gazpoucho
 - Scrapy

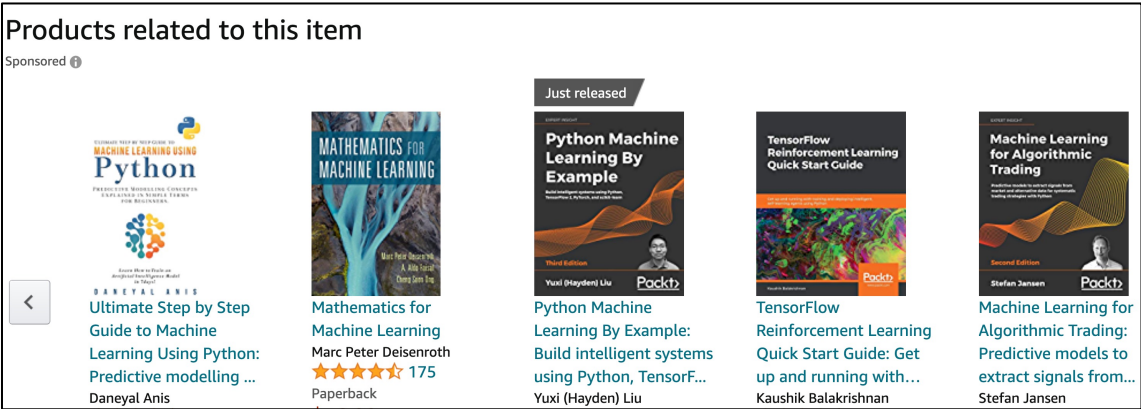
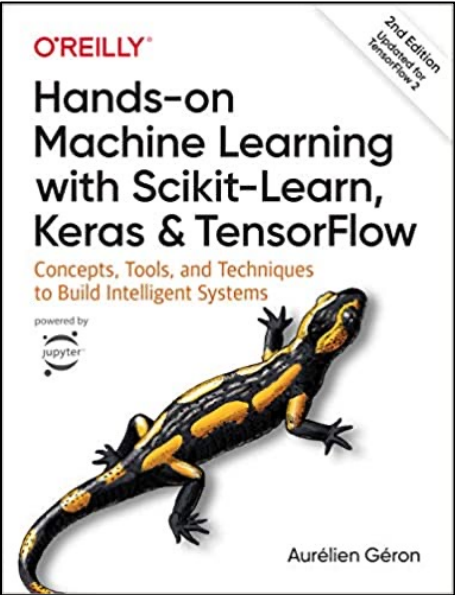


MACHINE LEARNING

ML / Case Study 1: Recommendations



Over 75% of what people watch comes from a recommendation Ranking



ML / Case Study 2: Classification



Likelihood How probable is the evidence given that our hypothesis is true?	Prior How probable was our hypothesis before observing the evidence?
$P(H e) = \frac{P(e H) P(H)}{P(e)}$	
Posterior How probable is our hypothesis given the observed evidence? (Not directly computable)	Marginal How probable is the new evidence under all possible hypotheses? $P(e) = \sum P(e H_i) P(H_i)$

Search for the Wreckage of Air France Flight AF 447

Lawrence D. Stone, Colleen M. Keller, Thomas M. Kratzke and Johan P. Strumpfer

Machine Learning: Tasks, Techniques, and Tools

■ Classification / Regression

- assign a label (classification) or numerical value (regression) to an unknown entity based on a set of features and known labels (or numerical values)

■ Clustering

- group a bunch of entities that share common features

■ Optimization

- from amongst a set of alternatives pick the “best” while balancing competing value metrics

■ Forecasting

- based on the past, predict the future

■ Recommendation

- based on prior behavior rank order candidate preferences

■ Association

- identify which items co-occur e.g., bread and peanut butter

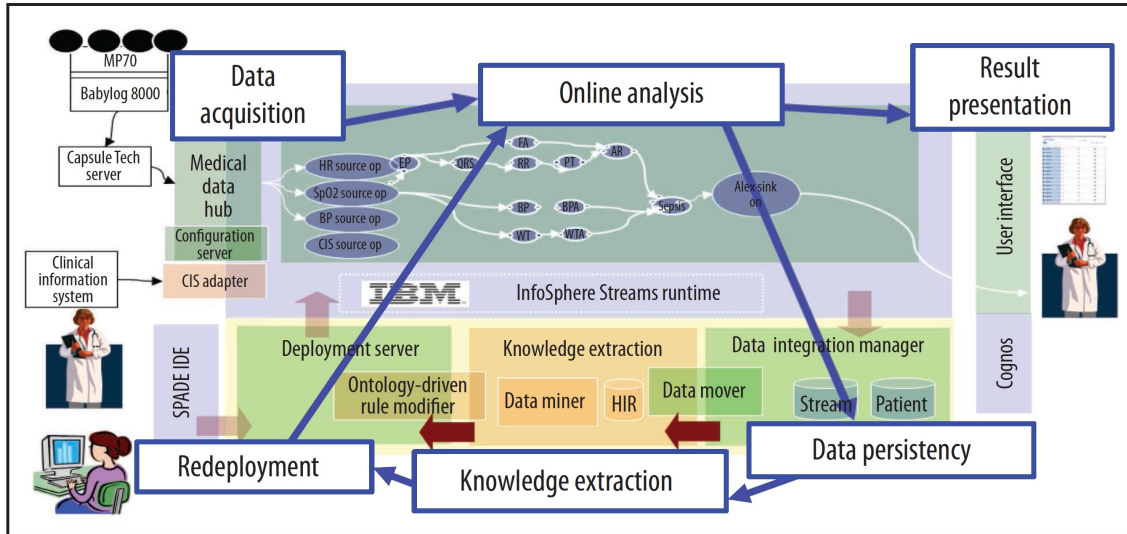
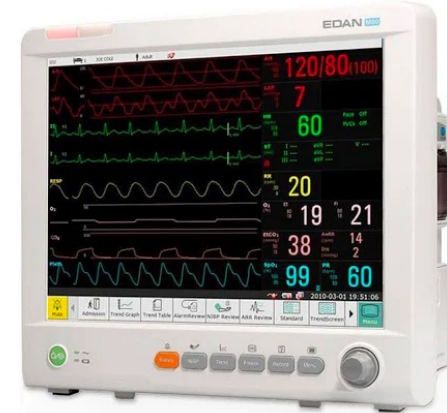
Tools:

- Scikit-Learn
- SparkML

BIG DATA

Big Data / Case Study 1: Project Artemis

About 20 percent of low-birth-weight babies develop an infection, and of those babies about 18 percent actually pass away.



40 beds gives rise to 5 TB / day

Synthesize the millions of data samples coming from the neonatal monitors to see if there were any notable patterns. ... when you look at more data and you look at it in a more granular sense, *not only is there more to be seen, but you actually see more.*

Tools for Big Data

- Hadoop, MapReduce
- Spark
- Cloud Platforms



MrJob



Spark
SQL

Spark
Streaming

MLlib
(machine
learning)

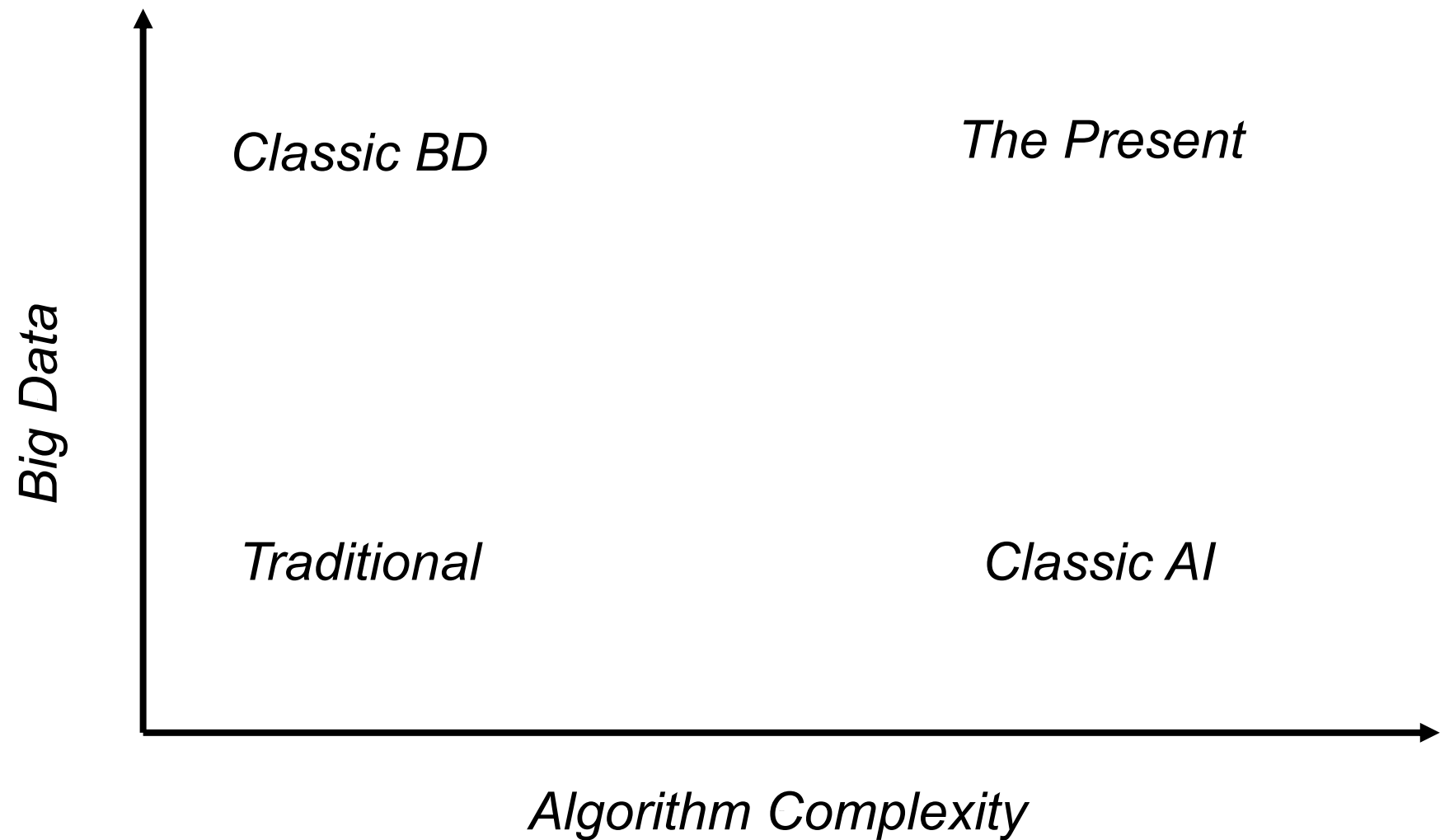
GraphX
(graph)

Apache Spark

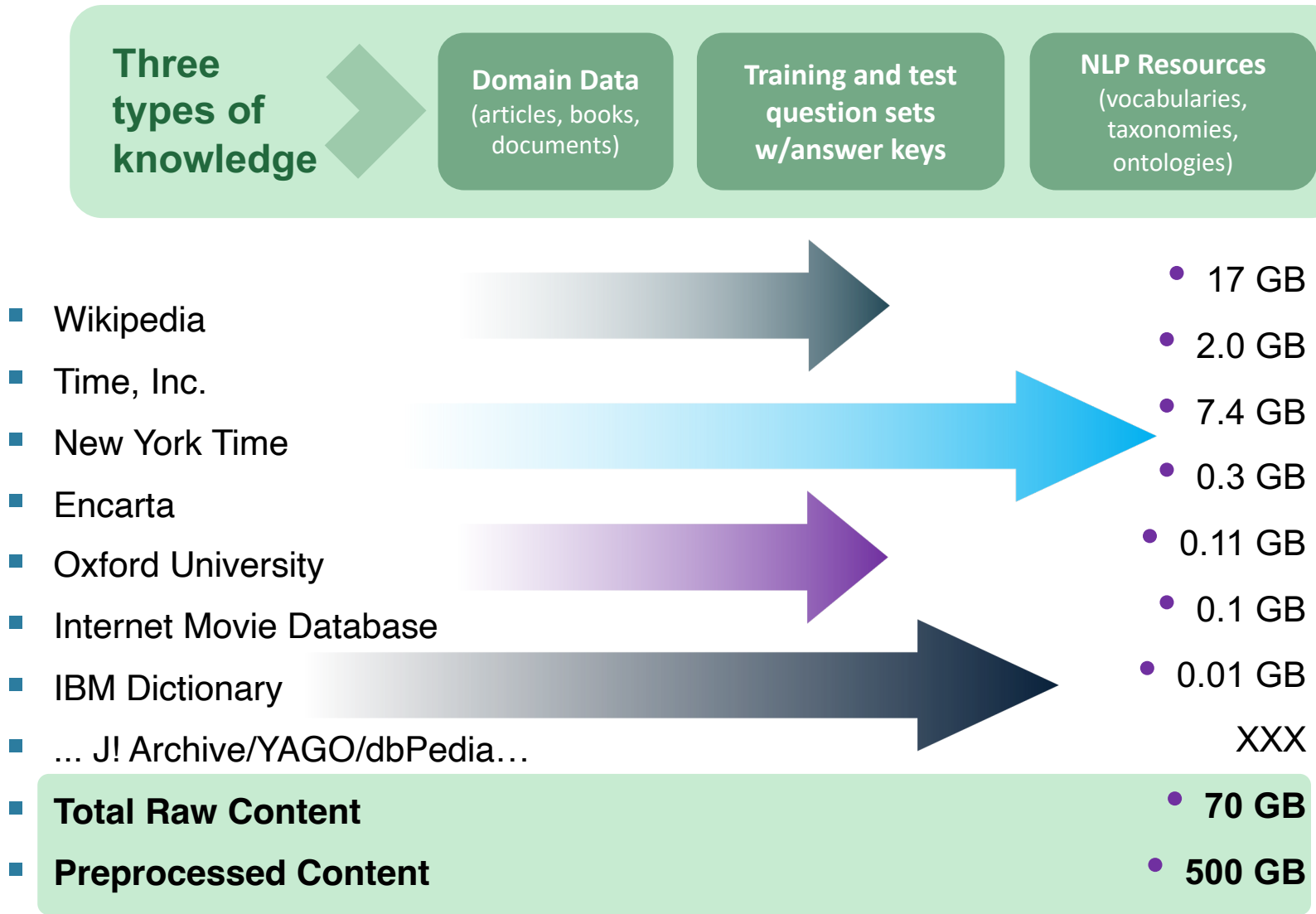


databricks

Data Vs. Algorithms



Big Data / Case Study 2: Cognitive Computing



The Pendulum may swing from Data back to Algorithms

The Future of AI Will Be About Less Data, Not More

**Harvard
Business
Review**

by H. James Wilson, Paul R. Daugherty, and Chase Davenport

Summary. Companies considering how to invest in AI capabilities should first understand that over the coming five years applications and machines will become less artificial and more intelligent. They will rely less on bottom-up big data and more on top-down reasoning that more closely resembles the way humans approach problems and tasks. This general reasoning ability will enable AI to be more broadly applied than ever, creating opportunities for early adopters even in businesses and activities to which it previously seemed unsuited. Current AI has lots of limitations. It especially has trouble handling “edge” cases—situations where little data exists for training. A driverless car that can handle crosswalks, pedestrians, and traffic has trouble processing anomalies like children dressed in unusual Halloween costumes, weaving across the street after dusk. Many systems are also easily stumped. The iPhone X’s facial recognition system doesn’t recognize “morning faces”—a user’s puffy, haggard look on first awakening. But in the coming years, systems that learn “top-down” from limited data will supplant these data-hungry bottom-up applications. They’ll have more efficient reasoning, ready expertise, common sense. Smart companies will place their bets on these more “human” systems.

STOPPED OCT 12

NOTES

- <https://www.forbes.com/sites/brentdykes/2016/03/31/data-storytelling-the-essential-data-science-skill-everyone-needs/?sh=34b4466252ad>
- <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/hal-varian-on-how-the-web-challenges-managers#>
- The ability to take data—to be able to understand it, to process it, to extract value from it, to visualize it, to communicate it—that's going to be a hugely important skill in the next decades, not only at the professional level but even at the educational level for elementary school kids, for high school kids, for college kids. Because now we really do have essentially free and ubiquitous data. So the complimentary scarce factor is the ability to understand that data and extract value from it.

MovieLens

MovieLens 100K Dataset

Stable benchmark dataset. 100,000 ratings from 1000 users on 1700 movies. Released 4/1998.

- [README.txt](#)
- [ml-100k.zip](#) (size: 5 MB, [checksum](#))
- [Index of unzipped files](#)

Permalink: <http://grouplens.org/datasets/movielens/100k/>

Data

Last 19 fields of u.item indicate the genre. Further details in the README file

File	Fields
u.data (10000)	user id movie id rating timestamp
u.item (1682)	movie id movie title release date video release date IMDB URL unknown Action Adventure Animation Children's Comedy Crime Documentary Drama Fantasy Film-Noir Horror Musical Mystery Romance Sci-Fi Thriller War Western
u.User (102625)	user id age gender occupation zip code

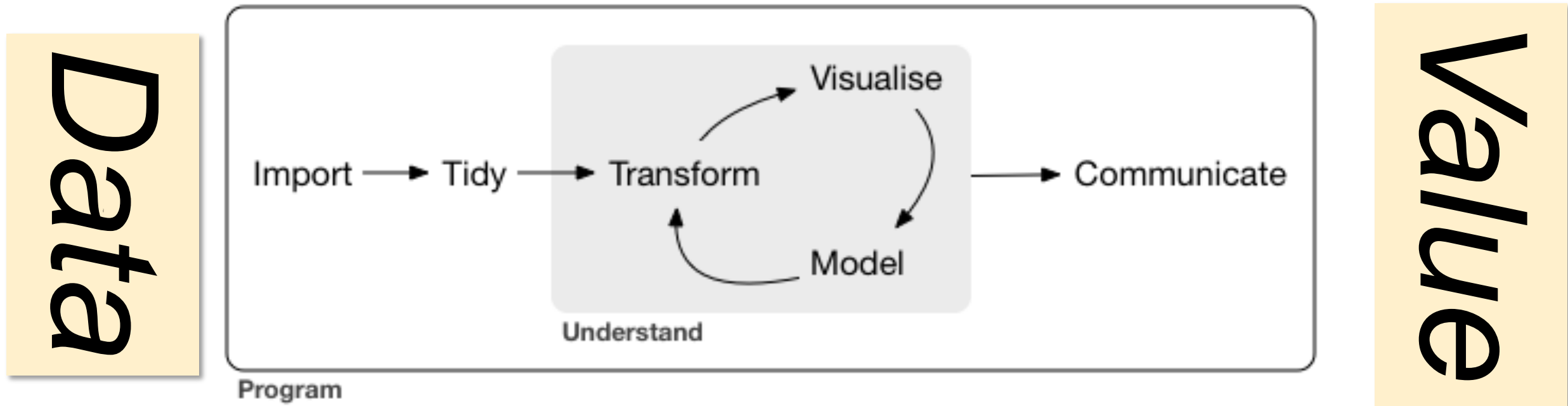
Question: Given the above files what questions could we ask about this data?

File	Fields
u.data (10000)	user id movie id rating timestamp
u.item (1682)	movie id movie title release date video release date IMDB URL unknown Action Adventure Animation Children's Comedy Crime Documentary Drama Fantasy Film-Noir Horror Musical Mystery Romance Sci-Fi Thriller War Western
u.User (102625)	user id age gender occupation zip code

Question: Given the above files (fields) how would you conduct the analysis to answer the question:
How do men and women rate movies differently? Try to answer the following questions on paper:

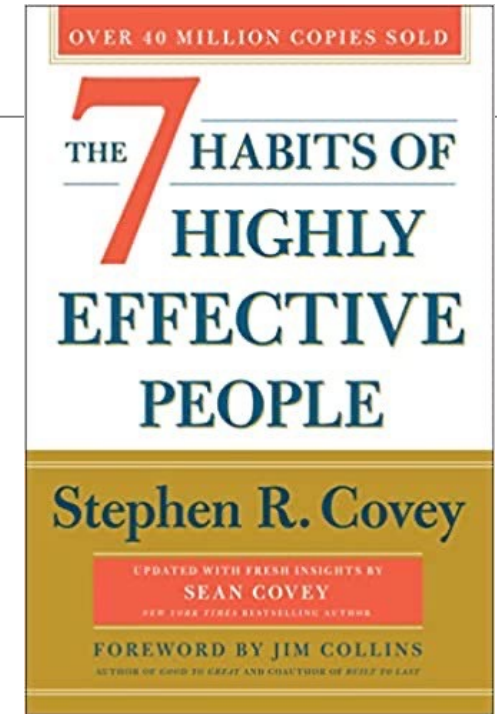
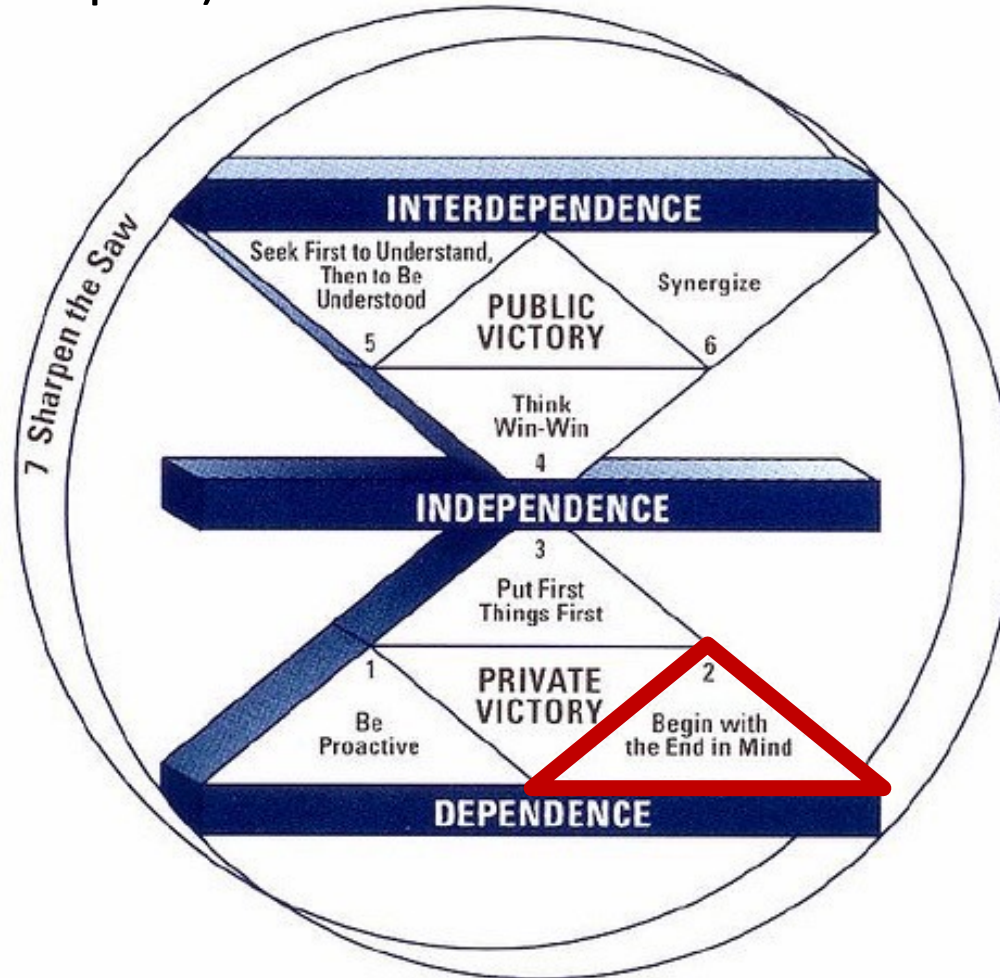
1. How would you combine the data in the 3 tables into a single table? What fields do you need?
2. How would you analyze the combined table?
3. What would be the data format (tabular format) of the final analysis?
4. How would you visually convey your final analysis?

Data Science: Extracting value from data



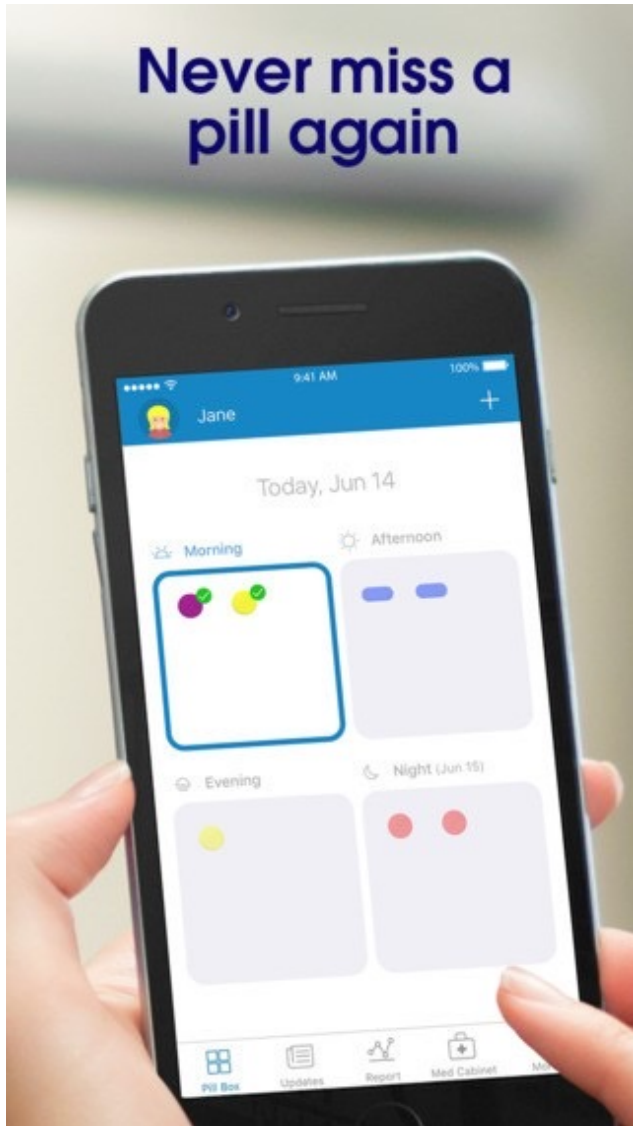
Value Proposition (end-to-end)

- Pain Point
- Problem Formulation (measure the pain)
- Solution Development
- Deployment
- Evaluation (reduction in pain)
- Maintenance / Sustainability

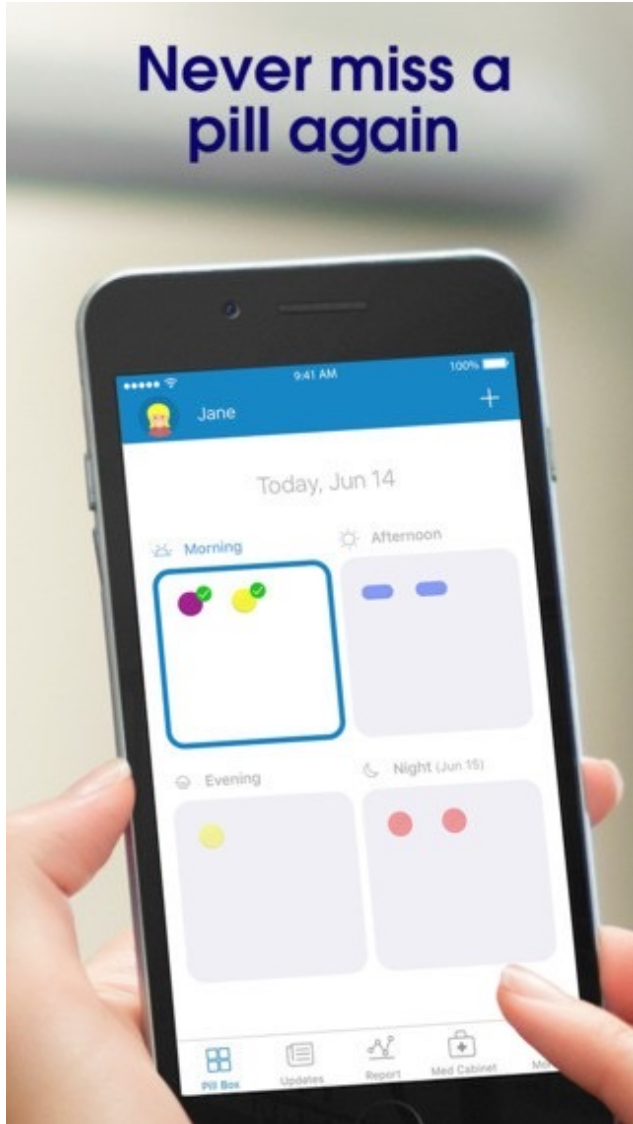


VALUE PROPOSITION

Solve the "problem" ... (it is NOT about technology)



Solve the "problem" ... (it is NOT about technology)



Business Problem

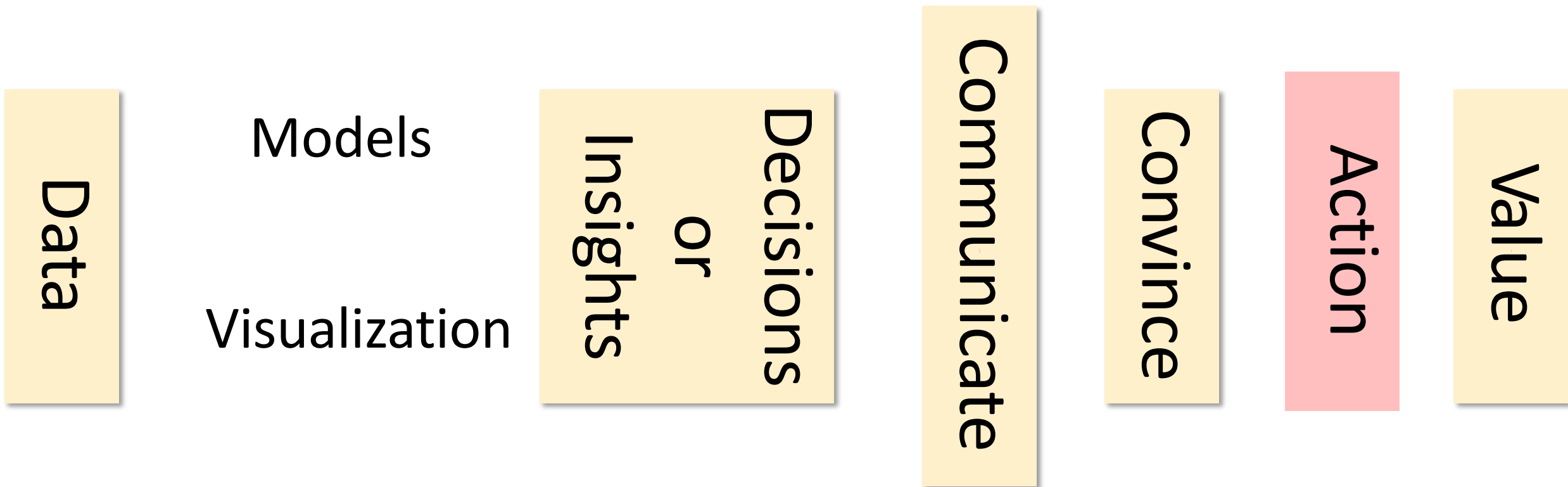
Business Solution

DA/ML Problem

DA/ML Solution



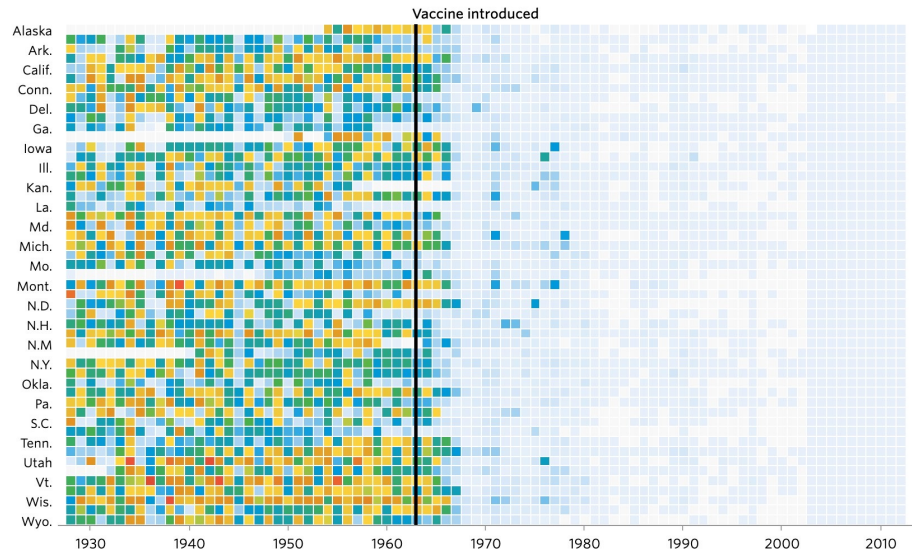
Analytics to Action



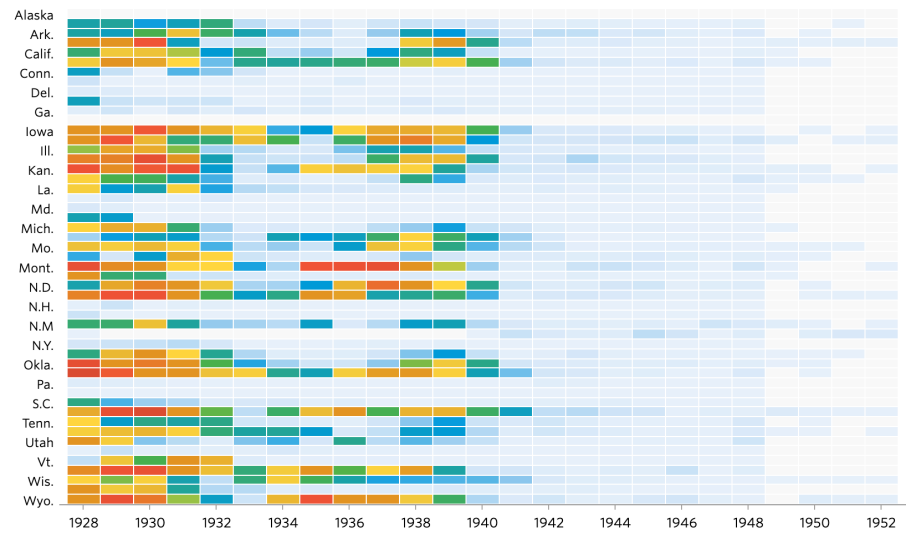
Battling Infectious Diseases in the 20th Century: The Impact of Vaccines

<http://graphics.wsj.com/infectious-diseases-and-vaccines/>

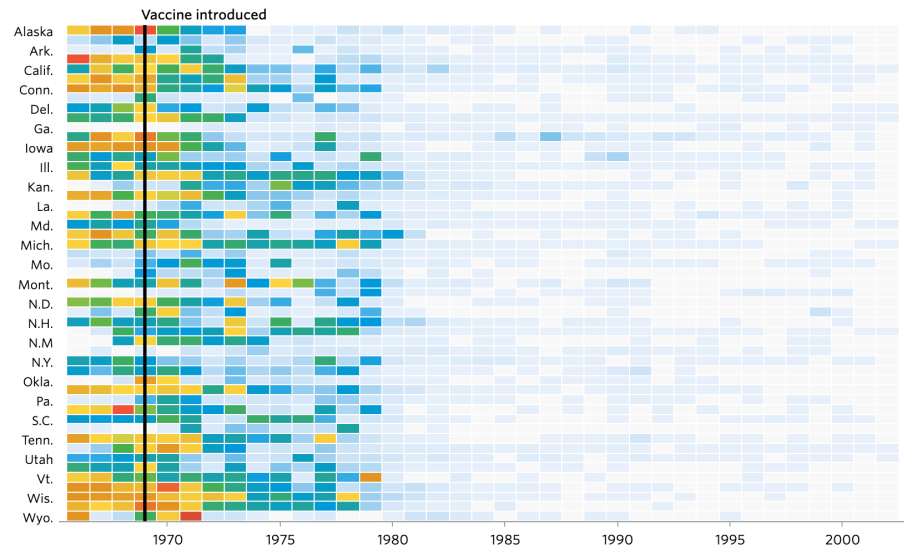
Measles



Smallpox



Rubella



Polio

